

1 Sylvester-equation-based solver

1.1 Sylvester reformulation of the posterior uncertainty term.

For online prediction, the most expensive part of the predictive variance is the posterior uncertainty correction

$$K_{*u}K_{uu}^{-1}SK_{uu}^{-1}K_{u*}.$$

Using the online sufficient-statistics form, let

$$\Lambda_n = \Lambda_0 + \frac{1}{\sigma^2} \sum_{j=1}^n A_j^\top A_j, \quad S_n = \Lambda_n^{-1}. \quad (1)$$

Define

$$q_* := K_{uu}^{-1}K_{u*}. \quad (2)$$

Then the posterior uncertainty term becomes

$$K_{*u}K_{uu}^{-1}S_nK_{uu}^{-1}K_{u*} = q_*^\top S_n q_* = q_*^\top \Lambda_n^{-1} q_*. \quad (3)$$

Now recall that for each temporal block j , the online projection matrix has the Kronecker form

$$A_j = T_j \otimes C, \quad (4)$$

where

$$T_j := K_{fu,j}^{(t)}(K_{uu,j}^{(t)})^{-1}, \quad C := K_{XZ}^{(s)}(K_{ZZ}^{(s)})^{-1}. \quad (5)$$

Hence

$$A_j^\top A_j = (T_j^\top T_j) \otimes (C^\top C). \quad (6)$$

Assuming the initial precision is separable,

$$\Lambda_0 = K_t^{-1} \otimes K_s^{-1}, \quad (7)$$

with $K_s := K_{ZZ}^{(s)}$, it follows that

$$\Lambda_n = K_t^{-1} \otimes K_s^{-1} + B_n \otimes G, \quad (8)$$

where

$$B_n := \frac{1}{\sigma^2} \sum_{j=1}^n T_j^\top T_j, \quad G := C^\top C. \quad (9)$$

To evaluate $q_*^\top \Lambda_n^{-1} q_*$ without explicitly forming the full matrix S_n , consider the linear

system

$$\Lambda_n z = q_*. \quad (10)$$

Let

$$q_* = \text{vec}(Q), \quad z = \text{vec}(Z), \quad (11)$$

with $Q, Z \in \mathbb{R}^{M_s \times M_t}$. Using the identity

$$(A \otimes B) \text{vec}(X) = \text{vec}(BXA^\top),$$

equation (8) yields

$$K_s^{-1} Z K_t^{-T} + G Z B_n^\top = Q. \quad (12)$$

If K_t , K_s , B_n , and G are symmetric, this simplifies to

$$K_s^{-1} Z K_t^{-1} + G Z B_n = Q. \quad (13)$$

Thus, the large $(M_t M_s)$ -dimensional linear system is reduced to a Sylvester-type matrix equation of size $M_s \times M_t$.

Next, introduce the whitened variables

$$\tilde{Z} := K_s^{-1/2} Z K_t^{-1/2}, \quad \tilde{Q} := K_s^{1/2} Q K_t^{1/2}, \quad (14)$$

and

$$\tilde{G} := K_s^{1/2} G K_s^{1/2}, \quad \tilde{B} := K_t^{1/2} B_n K_t^{1/2}. \quad (15)$$

Then (13) becomes

$$\tilde{Z} + \tilde{G} \tilde{Z} \tilde{B} = \tilde{Q}. \quad (16)$$

Let

$$\tilde{G} = U_s \Gamma U_s^\top, \quad \tilde{B} = U_t D_t U_t^\top, \quad (17)$$

where

$$\Gamma = \text{diag}(\gamma_1, \dots, \gamma_{M_s}), \quad D_t = \text{diag}(\beta_1, \dots, \beta_{M_t}). \quad (18)$$

Define the rotated variables

$$\hat{Z} := U_s^\top \tilde{Z} U_t, \quad \hat{Q} := U_s^\top \tilde{Q} U_t. \quad (19)$$

Substituting these into (16) gives

$$\hat{Z} + \Gamma \hat{Z} D_t = \hat{Q}. \quad (20)$$

Since Γ and D_t are diagonal, the equation decouples entrywise:

$$(1 + \gamma_i \beta_j) \hat{Z}_{ij} = \hat{Q}_{ij}, \quad (21)$$

and therefore

$$\hat{Z}_{ij} = \frac{\hat{Q}_{ij}}{1 + \gamma_i \beta_j}. \quad (22)$$

Finally, since $z = \Lambda_n^{-1} q_*$, we obtain

$$q_*^\top \Lambda_n^{-1} q_* = q_*^\top z = \text{vec}(Q)^\top \text{vec}(Z) = \langle Q, Z \rangle_F. \quad (23)$$

Hence the posterior uncertainty term in the predictive variance can be evaluated exactly as

$$K_{*u} K_{uu}^{-1} S_n K_{uu}^{-1} K_{u*} = \langle Q, Z \rangle_F, \quad (24)$$

where Z is obtained from the Sylvester equation (13), without explicitly forming the full posterior covariance

$$S_n \in \mathbb{R}^{(M_t M_s) \times (M_t M_s)}.$$

1.2 Interpretation.

The above derivation is an exact reformulation rather than a posterior approximation. Its computational gain comes from three structural facts: (i) the online block matrix A_j is Kronecker-separable, (ii) the spatial factor $C = K_{XZ}^{(s)} (K_{ZZ}^{(s)})^{-1}$ is fixed across time, and (iii) the initial precision Λ_0 is separable. Consequently, the large system

$$\Lambda_n z = q_*$$

is replaced by an $M_s \times M_t$ Sylvester equation, avoiding direct inversion or Cholesky factorization of the full $(M_t M_s) \times (M_t M_s)$ covariance matrix.

1.3 Complexity remark.

The computational gain of the Sylvester reformulation can be quantified as follows. If one directly forms and factorizes the full posterior covariance (or precision) over the mixed inducing variables, whose dimension is $M_t M_s$, then a dense direct solve typically costs

$$O((M_t M_s)^3).$$

By contrast, under the Kronecker-separable online update

$$\Lambda_n = K_t^{-1} \otimes K_s^{-1} + B_n \otimes G,$$

the quadratic form $q_*^\top \Lambda_n^{-1} q_*$ is reduced to solving the Sylvester equation

$$K_s^{-1} Z K_t^{-1} + G Z B_n = Q.$$

This can be carried out by:

1. forming or updating the temporal Gram-type matrix $B_n \in \mathbb{R}^{M_t \times M_t}$, costing $O(M_t^3)$ in the worst case under dense linear algebra;
2. eigendecomposing the whitened spatial matrix $\tilde{G} \in \mathbb{R}^{M_s \times M_s}$, costing $O(M_s^3)$;
3. eigendecomposing the whitened temporal matrix $\tilde{B} \in \mathbb{R}^{M_t \times M_t}$, costing $O(M_t^3)$;
4. applying the two basis transforms and the entrywise solve

$$\hat{Z}_{ij} = \frac{\hat{Q}_{ij}}{1 + \gamma_i \beta_j},$$

which costs $O(M_t^2 M_s + M_t M_s^2)$.

Therefore, the overall cost is reduced from

$$O((M_t M_s)^3)$$

to

$$O(M_t^3 + M_s^3 + M_t^2 M_s + M_t M_s^2),$$

up to constant factors and depending on whether some eigendecompositions can be cached across online steps.

In particular, when the spatial factor is fixed across time, the eigendecomposition associated with the spatial side may be precomputed once and reused, so the amortized online cost is dominated by the temporal update and the two-sided matrix transforms.